

FPGA Implementation of QPSK modulator by using Hardware Co-simulation

Preksha R. Kolankar¹, Swati V. Sakhare²

¹M.E.Student, Department of Electronics & Telecommunications Amravati University, Amravati, MS, India

²Asst. Prof, Departments of Electronics & Telecommunications Amravati University, Amravati, MS, India

Abstract:- The modulators are the basic requirement of communication systems they are designed to reduce the channel distortion & to use in RF communication hence many type of carrier modulation techniques has been already proposed according to channel properties & data rate of the system. QPSK (Quadrature Phase Shift Keying) is one of the modulation schemes used in wireless communication system due to its ability to transmit twice the data rate for a given bandwidth. The QPSK is the most often used scheme since it does not suffer from BER (Bit Error rate) degradation while the bandwidth efficiency is increased. It is very popular in Satellite communication. As the design of complex mathematical models such as QPSK modulator in 'pure HDL' is very difficult and costly; it requires from designer many additional skills and is time-consuming.

To overcome these types of difficulties, the proposed QPSK modulator can be implemented on FPGA by using the concept of hardware co-simulation at Low power. In this process, QPSK modulator is simulated with Xilinx System Generator Simulink software and later on it is converted in Very high speed integrated circuit Hardware Descriptive Language to implement it on FPGA. Along with the co-simulation, power of the proposed QPSK modulator can be minimized than conventional QPSK modulator. As a conclusion, the proposed architecture will not only able to operate on co-simulation platform but at the same time it will significantly consume less operational power.

Keywords:- QPSK, VHDL, FPGA, Xilinx

I. INTRODUCTION

Modulation is the method of transmitting data signal over carrier signal to minimize the distortion in channel. In Digital modulation, the digital information is transferred as a series of ones (1) and zeros(0) as input signal and amplitude, frequency or phase of carrier signal is shifted according to digital input, so this technique is also known as shift keying. But in digital modulation, instead of varying the amplitude or the frequency of the carrier signal, it is preferred to vary the phase because it offers better protection in transmitting signals.

Different PSKs can be obtained by M-ary PSK, where M is the no. of states or no. of phase shifts which is depend upon the no. of signals are combined for modulation. In QPSK two signals are combined for modulation. BER of QPSK is better than higher order PSK signals such as 8-PSK, 16-QAM, 32-QAM etc. which are easily affected by noise. At higher order PSK, larger bandwidth is require for higher data transfer rate and consume more power, whereas QPSK is more bandwidth as well as power efficient. There are many applications where QPSK modulator is used, out of which few are of battery operated devices such as Bluetooth, TDMA cellular communication, Medical Implant Communication Services (MICS) etc. therefore it is necessary to minimize the power consumption of these devices so that the battery will last for longer time. It can be done by reducing the speed of operation.

II. QPSK MODULATOR

QPSK (Quadrature Phase Shift Keying) is one of the modulation schemes used in wireless communication system due to its ability to transmit twice the data rate for a given bandwidth. The mathematical analysis shows that QPSK can be used either to double the data rate compared with a BPSK system while maintaining the same bandwidth of the signal, or to maintain the data-rate of BPSK but halving the bandwidth needed. In this latter case, the BER of QPSK is exactly the same as the BER of BPSK. The QPSK modulator illustrated in fig 1. The binary sequence is separated by the serial-to-parallel converter into odd-bit-sequence for I channel and even bit- sequence for Q channel. Table 1 represents the QPSK signal coordinates with different phase according to input bits.

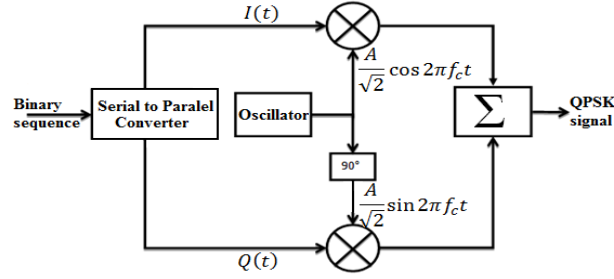


Figure 1: The QPSK Modulator

Table 1: QPSK Signal Coordinates

Digit	Phase θ_i	$s_{i1} = \sqrt{E} \cos \theta_i$	$s_{i2} = \sqrt{E} \sin \theta_i$
11	$\frac{\pi}{4}$	$+\sqrt{\frac{E}{2}}$	$+\sqrt{\frac{E}{2}}$
01	$\frac{3\pi}{4}$	$-\sqrt{\frac{E}{2}}$	$+\sqrt{\frac{E}{2}}$
00	$-\frac{3\pi}{4}$	$-\sqrt{\frac{E}{2}}$	$-\sqrt{\frac{E}{2}}$
10	$-\frac{\pi}{4}$	$+\sqrt{\frac{E}{2}}$	$-\sqrt{\frac{E}{2}}$

III. QPSK MODULATOR IN SIMULINK

The Simulink block set contains: The random source and rounding function blocks which generate the binary sequence or the modulating signal.

The binary sequence is separated into an odd-bit sequence I and an even-bit-sequence Q with the help of the sample-and-hold block. The Sample and Hold block acquires the input when it receives a trigger event at the trigger port (rising, falling or either edge). The block then holds the output at the acquired input value until the next triggering event occurs. The odd bits are acquired at the rising edge triggers: when the trigger input rises from a negative value or zero to a positive value and the even-bits are acquired at the falling edge triggers: when the trigger input falls from a positive value or zero to a negative value. Because the I sequence is read first, it has been delayed with the Transport Delay block in order to be synchronized with the Q sequence.

The cosine wave block generating a cosine waveform and with the help of the gain block, a cosine with 180° phase difference for the I-channel, and for the Q-channel: the sine wave block which generates a sinus waveform and with the help of the gain block, a sinus with phase difference of 180°.

The switch blocks which will choose between the first or third output depending on the value of the second input. In the case of the I-channel, if the second input is “1”, the output value will be cosine, but if the second input is “0”, the output will be –cosine and for the Q-channel, if the second input is “1”, the output value will be sin, but if the second input is “0”, the output will be –sin. The sum block combines the two modulated signals into the QPSK signal. Figure 3 illustrates the output waveforms.

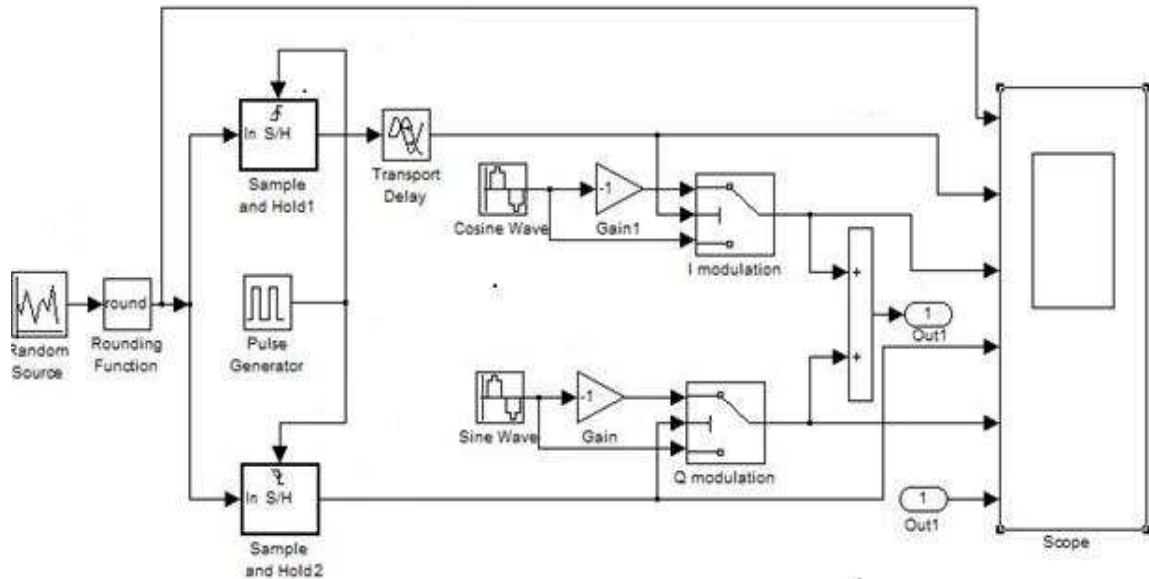


Figure2: QPSK Modulator in the Simulink environment.

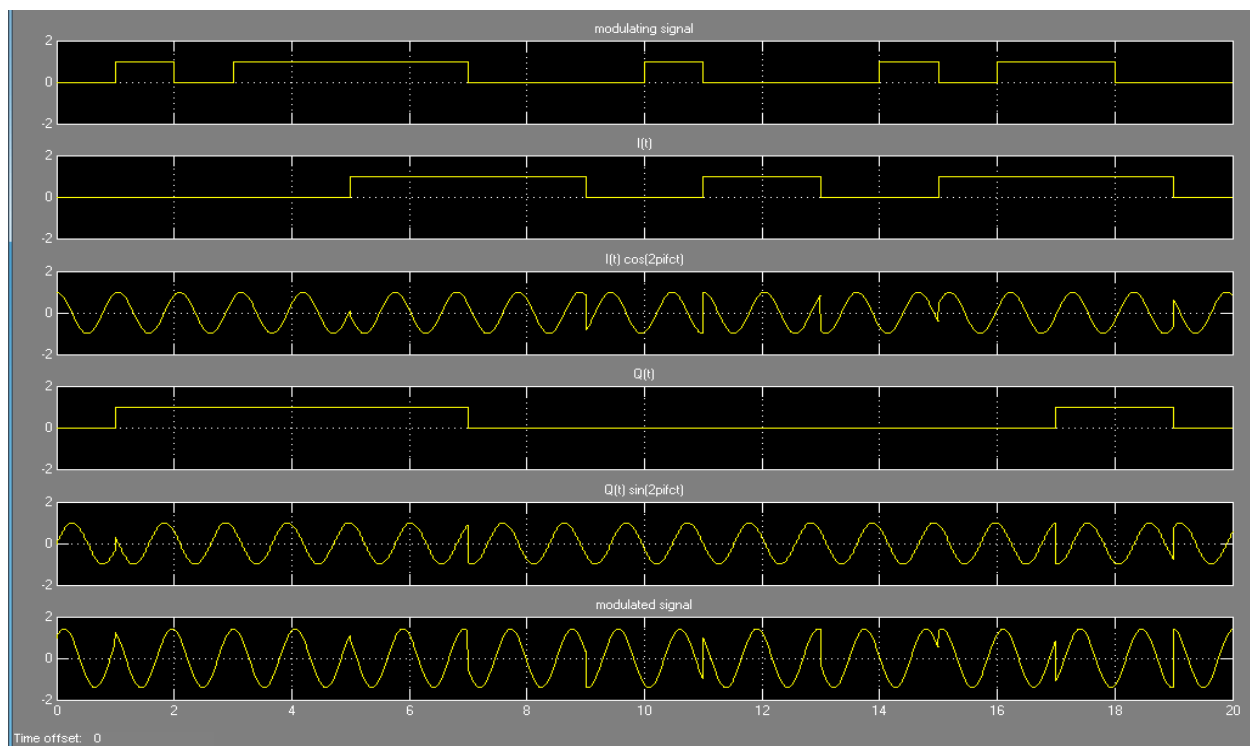


Figure 3: Output Waveforms of QPSK Modulator

QPSK Modulator in system Generator

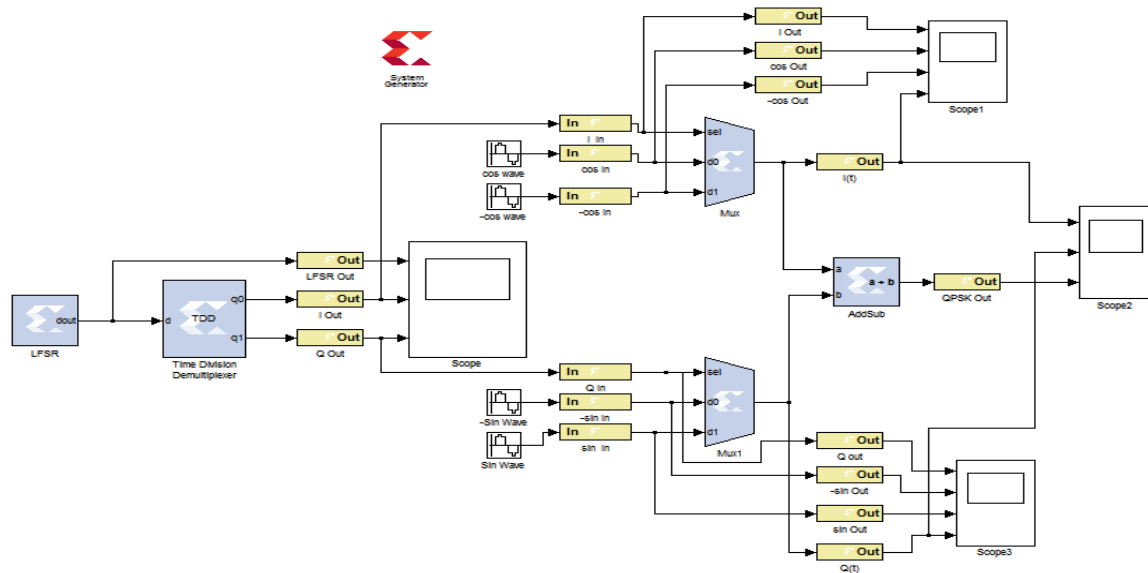


Figure 4. QPSK Modulator in System Generator.

System Generator is a digital signal processing design tool from Xilinx. It is based on the Matlab/Simulink environment used for FPGA design. Designs are made in the Simulink environment using a Xilinx specific blockset. All implementation steps, including synthesis, place and route are automatically performed to generate an FPGA programming file. Fig.4 illustrates the implementation of a QPSK Modulator using System Generator tools in Simulink.

The System Generator Blockset contains: gateway in blocks: It is used for giving inputs to the Xilinx portion of the Simulink design; gateway out blocks: It is output from the Xilinx portion of the Simulink design;

LFSR block: It implements a Linear Feedback Shift Register which is used to create the binary sequence or modulating signal.

Time Division Demultiplexer block: It accepts input serially and presents it to multiple outputs to a slower rate. In another word, this block divides the modulating signal into the odd-sequence I and even-sequence Q. The division take places inside the FPGA Fig.5 illustrate the waveforms from the first scope in System Generator. In order to see any signal on the scope, the signal must be passed through a gateway out.

Mux block: It implements a multiplexer. It has one select input and a configurable number of data inputs that can be defined by the user. The d0 and d1 inputs of the mux represent cosine and -cosine for the I sequence and sine and -sine for the Q sequence. The sel input selects between d0 and d1 depending on the odd-sequence for mux1 and the even-sequence for mux. The corresponding signals are illustrated in fig.6 and 7.

Add Sub block: It implements an adder (fig 8).

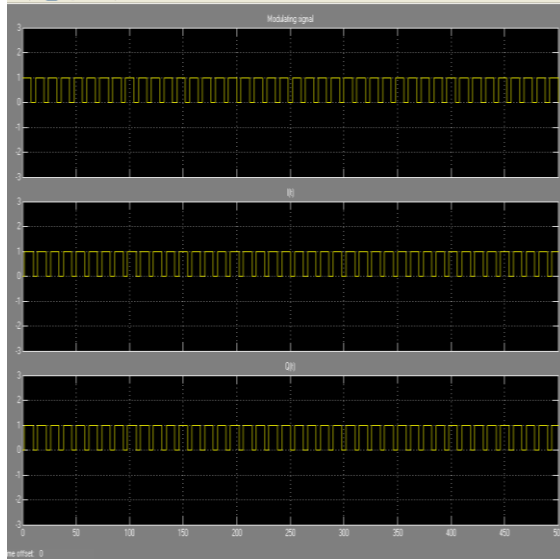


Fig 5 : a) Modulating Signal b) $I(t)$ c) $Q(t)$

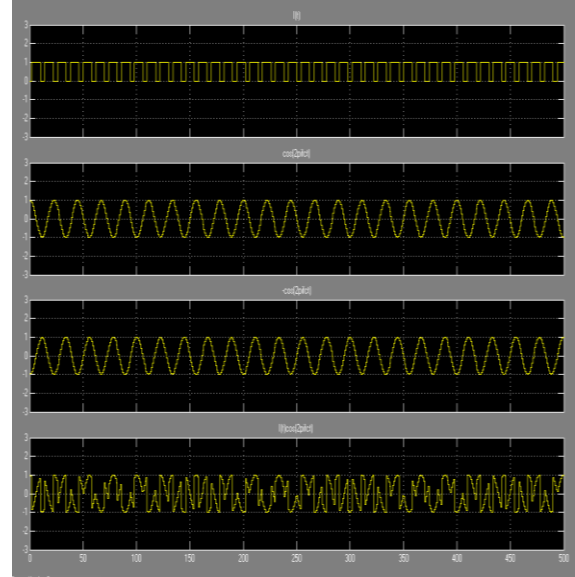


Fig 6: a) $I(t)$ b) $\cos(2pifct)$ c) $-\cos(2pifct)$ d) $I(t)\cos(2pifct)$

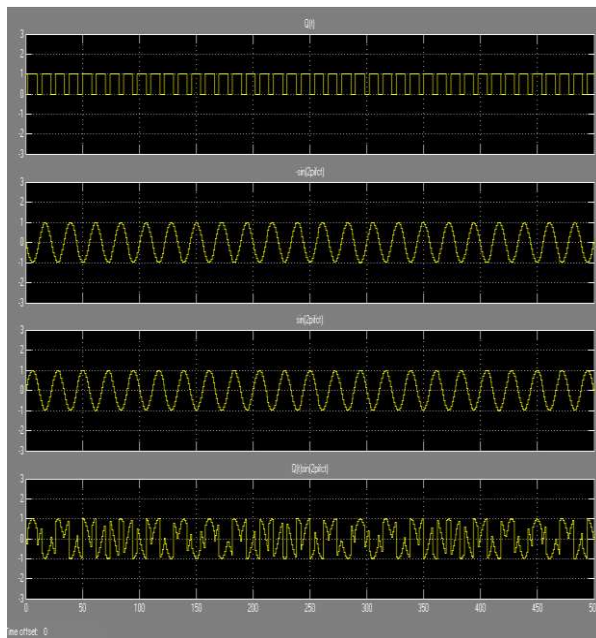


Fig 7: a) $Q(t)$ b) $-\sin(2pifct)$ c) $\sin(2pifct)$ d) $I(t)\cos(2pifct)$

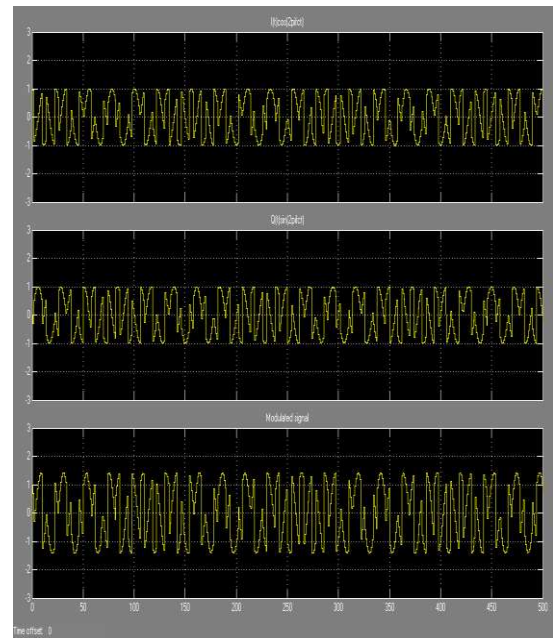


Fig 8: a) $I(t)\cos(2pifct)$ b) $Q(t)\sin(2pifct)$ c) Modulated signal

System Generator

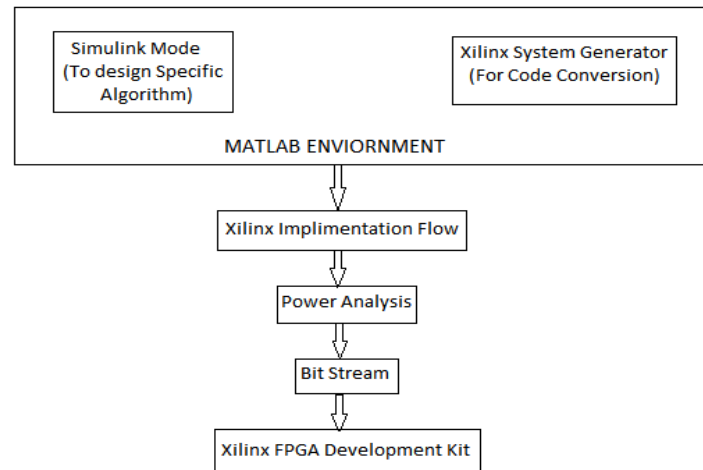


Figure 9: Block Diagram of System Generator

When System Generator compiles, it automatically wires the imported module and associated files into the surrounding netlist. Xilinx system generator is a very useful tool for developing computer vision algorithms. It could be described as a timely, advantageous option for developing in a much more comfortable way than that permitted by VHDL or Verilog hardware description languages (HDLs).

Matlab-Simulink is an environment for multidomain simulation and Model-Based Design for dynamic and embedded systems. Matlab-Simulink is used in this application as the high level development tool in the design process. Xilinx System Generator is a system-level modeling tool from Xilinx that facilitates FPGA hardware design. It extends Simulink in many ways to provide a modeling environment well suited for hardware design. The result can be synthesized to Xilinx FPGA technology using ISE tools. All of the downstream FPGA implementation steps including synthesis and place and route are automatically performed to generate an FPGA programming file. It also provides a system integration platform for the design of DSP FPGAs that allows the RTL, Simulink, MATLAB and C/C++ components of a DSP system to come together in a single simulation and implementation environment. System Generator supports a box block that allows RTL to be imported into Simulink and co-simulated with either ModelSim or Xilinx ISE Simulator.

Power Analysis:

After implementing the QPSK modulator on System generator, and by configuring this module on Xilinx ISE, all steps, including synthesis, place and route are automatically performed to generate an FPGA programming file. The power analysis is done by X-Power Analyser(XPA) tool which performs power estimation at post implementation stages. It is the most accurate tool since it can read from the exact logic and routing resources used for design. For further stage, the power analyzer tool provided by Xilinx is PlanAhead RTL power estimator. PlanAhead reads the HDL code from a design to estimate the resources needed and reports the estimated power from a statistical analysis of the activity of each resource. The text power report generated in Xilinx ISE gives the value of power consumed by the QPSK modulator as 38.06 mW.

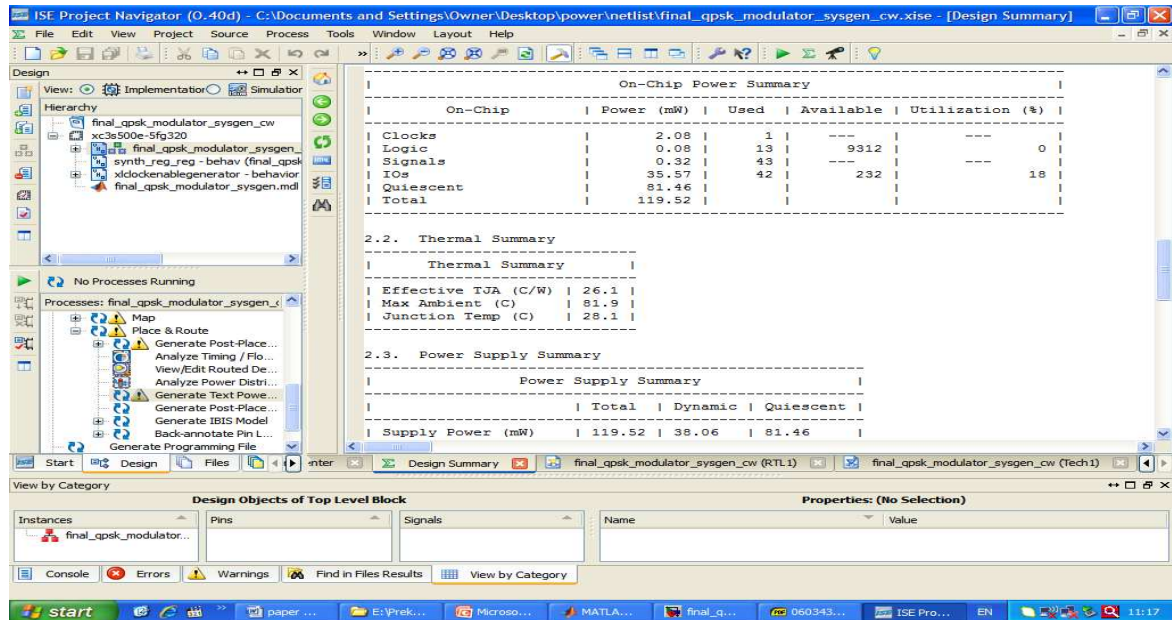


Figure 10: Power Report of QPSK Modulator

Hardware-Software Co-simulation

System Generator the use of simulates automatically launching an HDL simulator, generating additional HDL as needed (analogous to an HDL testbench), compiling HDL, scheduling simulation events, and handling the exchange of data between the Simulink and the HDL simulator. This is called HDL co-simulation.

System Generator provides hardware co-simulation, making it possible to incorporate a design running in an FPGA directly into a Simulink simulation. "Hardware Co-Simulation" compilation targets automatically create a bit-stream and associate it to a block as shown in given figure.

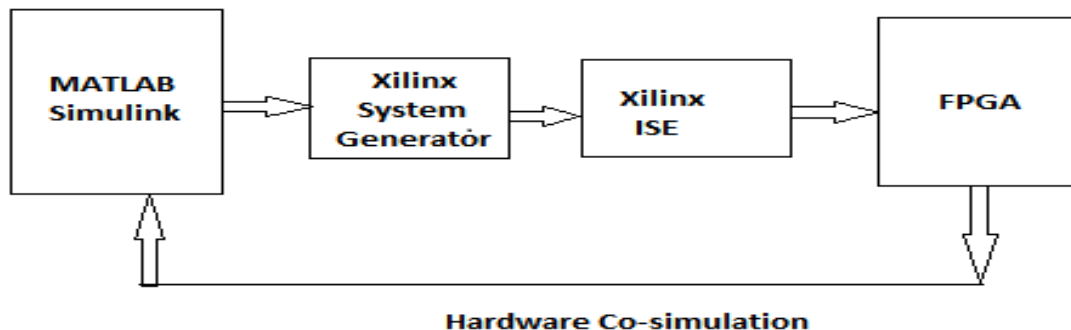


Figure 11: FPGA based Hardware-Software (HW-SW)co-simulation Environment

When the system design is simulated in Simulink, results for the compiled portion are calculated in actual FPGA hardware, often resulting in significantly faster simulation times while verifying the functional correctness of the hardware. System Generator for DSP supports Ethernet as well as JTAG communication between a hardware platform and Simulink. System Generator provides a generic interface that uses JTAG and a Xilinx programming cable (e.g., Parallel Cable IV or Platform Cable USB) to communicate with FPGA hardware. Fig shows the model with the JTAG-based hardware co-simulation block. Point-to-point Ethernet co-simulation provides a straightforward high-performance co-simulation environment using a direct, point-to-point Ethernet connection between a PC and FPGA platform. The optimization setting is for maximum clock speed. Xilinx system generator has a unique hardware in the loop co-simulation feature that allows designers to greatly accelerate simulation while simultaneously verifying the design in hardware.

The QPSK modulator in system generator is implemented on Spartan 3e board by using Hardware Software co-simulation through JTAG cable is shown in figure 12. This module will show output waveforms in Matlab itself, that means there is no need of using digital oscilloscope for observing waveforms, which can reduce the hardware cost efficiently.

